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1. (3) Concave mirror

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2. $u = -10 \text{ cm}$

$$m = -3$$

$$-3 = \frac{v}{-10}$$

$$30 \text{ cm} = v$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{1}{30} + \frac{1}{10}$$

$$= \frac{+1+3}{30}$$

$$= \frac{4}{30}$$

$$f = \frac{30}{4} = +7.5 \text{ cm}$$

(3) 30 cm, 7.5 cm

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3. (4) Both 2 and 3

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4. (i) Principal focus of spherical mirror - When light rays coming from infinity parallel to principal axis after reflection converge or appears to converge at a point behind the mirror is called principal focus.

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(ii) Centre of curvature of spherical mirror - The centre of the sphere from which the spherical mirror was a part of is known as centre of curvature.

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5. Given (convex lens)

(i) $m = \frac{-1}{5}$

$$v = + 18 \text{ cm}$$

$$m = \frac{v}{u}$$

$$\frac{-1}{5} = \frac{18}{u}$$

$$u = \frac{18 \times 5}{-1}$$

$$= -90 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{1}{18} + \frac{1}{90}$$

$$\frac{1}{f} = \frac{5+1}{90}$$

$$= \frac{6}{90}$$

$$f = \frac{90}{6} = +15 \text{ cm}$$

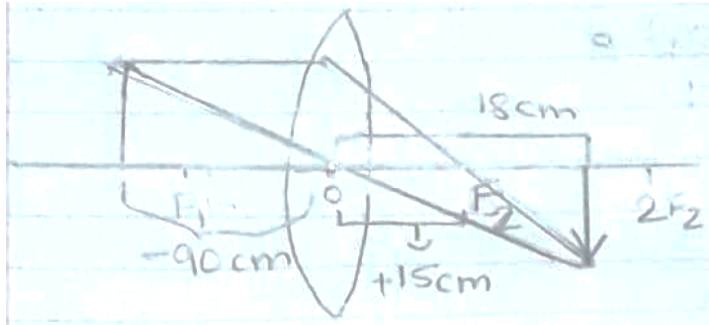
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(ii) Nature : The lens is convex lens convex lens is convergent.

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(iii)

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6. (i) Given (concave mirror)

$$h_o = +7 \text{ cm}$$

$$u = -27 \text{ cm}$$

$$f = -18 \text{ cm}$$

$$v = ?$$

$$h_i = ?$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{-1}{18} = \frac{1}{v} - \frac{1}{27}$$

$$\frac{1}{v} = \frac{1}{27} - \frac{1}{18}$$

$$= \frac{2-3}{54}$$

$$= \frac{-1}{54}$$

$$v = -54 \text{ cm}$$

$$m = \frac{-v}{u} = \frac{h_i}{h_o}$$

$$= \frac{-(-54)}{-27}$$

$$m = -2$$

$$-2 = \frac{h_i}{+7}$$

$$-14 \text{ cm} = h_i$$

Size : The image formed is 1 cm from the principal axis.

Nature : The image formed is real and inverted. The image formed is twice the size of the object.

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(ii) Given (convex mirror)

$$2f = +5 \text{ m}$$

$$u = 2 \text{ m}$$

$$v = ?$$

$$2f = +5 \text{ m}$$

$$f = \frac{+5}{2} \text{ m}$$

$$\frac{2}{5} = \frac{1}{v} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{2}{5} + \frac{1}{20}$$

$$= \frac{9}{20}$$

$$v = \frac{20}{9}$$

$$+2.22 \text{ m}$$

$$m = \frac{-v}{u}$$

$$= \frac{-20}{9} \div (-20)$$

$$= \frac{-\cancel{20}}{9} \times \frac{1}{\cancel{(-20)}} = \frac{1}{9}$$

Position : The image is formed +2.22 m behind the mirror.

Nature : The image formed is virtual and erect. The nature of the lens is diverging.

Size : The image formed is $\frac{1}{9}^{\text{th}}$ the size of the object.

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(iii) (a) The nature of this mirror is real and inverted. The mirror is converging.

(b) Concave mirror is used so that dentist get an enlarged image of the tooth.

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